Connected Multimedia Screens

Multimedia HandSet Reference Design

Overview

As the cordless technology leader for over two decades, DSP Group™ keeps up with worldwide telecommunications trends, enabling consumer electronics (CE) manufacturers to deliver the very best when it comes to cordless home phones. Recognizing rising consumer-wide demand for additional and more sophisticated mobile applications both in and outside the home, DSP Group is bringing the smart phone user experience to the cordless home phone via the interactive multimedia handset.

Combining industry-leading expertise and innovation, DSP Group has developed a highly advanced multimedia handset reference design. As part of the Company's connected multimedia screen family, the reference design enables cost-effective development of always-on multimedia handsets featuring an intuitive interface, user-friendly touch screen and virtual keyboard.

Changing the way users perceive their home phone, the next-generation handset provides immediate access to information and diverse applications including web browsing, widgets, music and e-mail. Comprised of a complete software set that includes an operating system, middleware and key array of applications, the reference design offers enhanced telephony and CAT-iq compliance. And by integrating the Google Android™ framework, the reference design results in not only increased freedom of choice, but also an enhanced user experience, by facilitating the discovery of applications and games offered by developers.

Highly portable, built on an open platform, and easily personalized, the DSP Group-designed multimedia handset is based on the Company's innovative XpandR™ II, a comprehensive chipset that integrates Wi-Fi and DECT. With an integrated processor, DSP, Wi-Fi and DECT baseband processor, analog front end, and software packages, XpandR II includes all of the components needed to develop a range of new and advanced portable and location-free multimedia products.

XpandR II is part of DSP Group's new-generation XpandR family of multimedia chipset solutions. Comprised of hardware and software development kits, the XpandR family integrates voice, data and video, reduces system bill of materials (BOM), and is suitable for mass-market adoption, while ensuring fast time to market.

By bringing Wi-Fi networking to the multimedia cordless handset, XpandR enables the development of a full range of value-added multimedia services. And by bringing DECT telephony to Wi-Fi networking, XpandR ensures that consumers will not have to compromise on voice quality.

Features

Software
- Linux OS with Google Android™ framework

User Interface and Applications
- Flash lite support
- Video playback (H.263 and 3GP formats)
- Remote management via web-based interface
- Accelerometer-based application rotations
- Calendar, contacts, call log and favorites
- Native Android web browser
- Widgets
  - Bundled home screen widgets including analog clock, music player, picture frame and search
- Application management

Google Applications
- Upload videos to YouTube
- Upload photos on Picasa

Processor
- ARM926EJ-S™ @ 240MHz processor
- 16-bit TeakLite™ DSP core for voice/audio processing functions at up to 160 MHz

Interfaces
- USB2.0 OTG
- SD/MMC/MS
- Keypad, LCD with touch screen

Display
- 3.5" TFT LCD display
- Multiple screen size support
- Portrait and landscape orientation

WLAN Connectivity
- Wi-Fi connectivity (802.11b/g) with maximal-ratio combining (MRC) operation for increased range and sensitivity
- Security: WEP, WPA, WPA2, TKIP and AES hardware acceleration
- QoS and low power: WMM and WMM-PS for improved battery life of Wi-Fi devices
- WPS for easy set-up and configuration
- Wi-Fi auto find (for easy Wi-Fi network setup)

DECT and CAT-iq
- 1.7GHz, 1.8GHz, 1.9GHz band
- GAP compliance, full CAT-iq 1.0/2.0 support
- Caller ID, call waiting CID

VoIP Software Suite
- Narrowband vocoders: G.711, G.723.1, G.726 G.729, G.729a
- Wide band vocoders: G.722
- SIP call signaling

Power Management
- 1350 mAh Li-Ion battery
- > 10 hours of DECT talk time
- > 5 hours of streaming content over Wi-Fi
- Up to 100 hours standby

Benefits

- Fast TTM
- Rapid new application integration
- Diverse, flexible offering
- Easy customization
- Enhanced user experience
- Low development costs

DSP GROUP
at home in your wireless world

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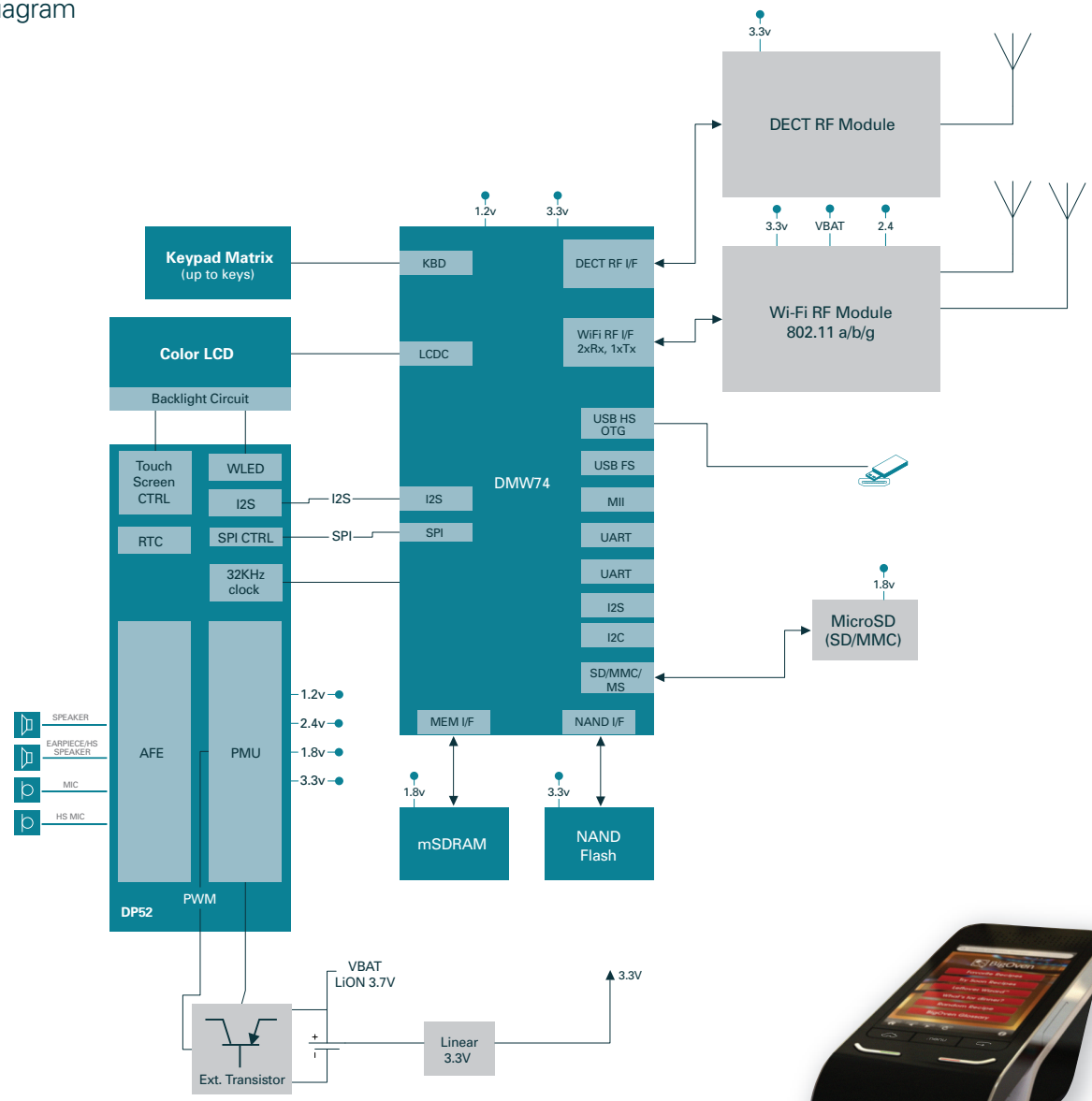
Block Diagram



DSP Group, Inc. (NASDAQ: DSPG) is a leading global provider of wireless chipset solutions for converged communications at home. Delivering system solutions that combine semiconductors and software with reference designs, DSP Group enables consumer electronics (CE) manufacturers to cost-effectively develop new revenue-generating applications with fast time to market. At the forefront of semiconductor innovation and operational excellence for over two decades, and with a dominant share of the wireless home telephony market, DSP Group provides a broad portfolio of wireless chipsets integrating DECT, Wi-Fi, PSTN and VoIP technologies with state-of-the-art application processors. Enabling converged voice, audio, video and data connectivity across diverse consumer products – from cordless and VoIP phones to home gateways and connected multimedia screens – DSP Group proactively partners with CE manufacturers to shape the future of converged communications at home.

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For more information on our products and services, visit our website at: **www.dspg.com** or contact us at: info@dspg.com

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